FED. SUP CLASS
5305

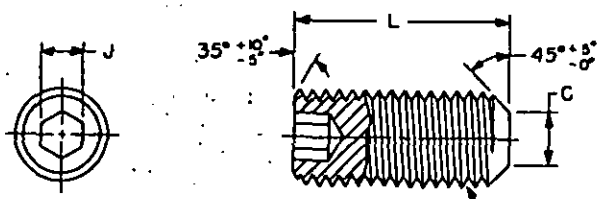


TABLE I

NOMINAL SIZE		.060	.086	.112	.138	.164	.190	.250	.3125	.375	.500
THREADS PER INCH (UNF-3A)		80	64	48	40	36	32	28	24	24	20
J SOCKET WIDTH ACROSS FLATS	NOM	.028	.035	.050	.062	.078	.094	.125	.156	.188	.250
C POINT DIAMETER	MAX	.033	.047	.061	.074	.087	.102	.132	.172	.212	.291
	MIN	.027	.039	.051	.064	.076	.088	.118	.156	.194	.270

L LENGTH	TOL	DASH NUMBER														
		PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	SELF LOCK	PLAIN	SELF LOCK	PLAIN	SELF LOCK	PLAIN	SELF LOCK	PLAIN	SELF LOCK
.062	±.010	1*														
.125			7*	16*	25*	35*		110*								
.188			8*	17*	26*	36*	48*	111		121*						
.250			9*	18*	27*	37*	49*	112	60*	122	73*	134				
.312				19*	28*	38*	50*	113	61*	123	74*	135	86*		98*	
.375					29*	39*	51*	114	62*	124	75*	136	87*	146	99*	156
.438								115*		125*						
.500						40*	52*	116	63*	126	76*	137	88*	147	100*	157
.562								117*		127*						
.625						41*	53*	118	64*	128	77*	138	89*	148	101*	158
.750	±.020					42*	54*	119	65*	129	78*	139	90*	149	102*	159
.875									66*	130	79*	140	91*	150	103*	160
1.000								120	67*	131	80*	141	92*	151	104*	161
1.250										132		142		152		162
1.500										133		143		153		163
1.750												144		154		164
2.000												145		155		165
2.500	±.031															166
3.000																167

* Inactive for new design after 29 DEC 1972.

(A) NOTES:

1. MATERIAL: Alloy steel in accordance with procurement specification, having a Rockwell Hardness of C45-53.
2. PROTECTIVE COATING: Cadmium plated in accordance with procurement specification.
3. THREADS: The threads shall be in accordance with MIL-S-7742.
4. DIMENSIONS: All dimensions are in inches, unless otherwise specified.
5. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51966-48.
6. SELF-LOCKING ELEMENT:
   (a) The self-locking element shall be in accordance with MIL-F-18240.
   (b) For design and usage limitations, see MS15981.
   (c) Location of effective locking area shall be in accordance with MS15981.
   (d) Maximum ring gage diameter that locking region of setscrew must pass thru freely or with finger pressure shall be the nominal size plus 0.010.
   (e) MIL-F-18240 doesn't establish torque requirements for lengths above the dashed line. Only a positive indication of torque is required.
7. Referenced documents shall be of the issue in effect on date of invitations for bid.
8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

(A) For changes see sheets 1 and 2.

REVIEWER: AV, IS, MI, MU
USER: AT, GL, MC, ME, OS, SH, YD

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 29 JAN 1964 REVISED (A) 29 DEC 1972

P.A. WC	TITLE	MILITARY STANDARD
Other Cust AS 82	SETSCREW-HEXAGON SOCKET, FLAT POINT, ALLOY STEEL, CADMIUM PLATED, UNF-3A, PLAIN AND SELF-LOCKING (A)	MS51966
PROCUREMENT SPECIFICATION FF-S-200	SUPERSEDES: MS51026, MS51028 and in part MS18065, BCTX2.6, BCTX2.7, BCTX2.13, BCTX2.15, BCUX1, BCUX3 and BCUX4.5	SHEET 1 OF 2

DD FORM 672-1 (Coordinated)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.

REVIEWER: AV, IS, MI, MU
USER: AT, GL, MC, ME, OS, SH, YD

INTERCHANGEABILITY

Screws covered by the dash numbers given in MS51026 and MS51028 are cancelled after 29 January 1964 and superseded by the screws covered on MS51966 having the same dash numbers. Cancelled screws may be used until existing stocks are depleted. Use only MS51966 screws for new design.

Screws covered by the dash numbers given in MS18065 in part are cancelled. Cancelled screws may be used until existing stocks are depleted. Use only the superseding screws for new design. Replacement shall be in accordance with Table II.

Ⓐ

TABLE II

CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING
MS18065	MS51966	MS18065	MS51966	MS18065	MS51966	MS18065	MS51966
DASH NUMBER		DASH NUMBER		DASH NUMBER		DASH NUMBER	
4	110	32	125	64	140	82	154
8	111	37	126	70	141	85	155
13	112	45	127	74	142	92	156
19	113	50	128	78	143	40	157
25	114	57	129	81	144	53	158
31	115	63	130	84	145	60	159
36	116	69	131	91	146	66	160
44	117	73	132	39	147	72	161
49	118	77	133	52	148	76	162
56	119	15	134	59	149	80	163
68	120	21	135	65	150	83	164
9	121	27	136	71	151	86	165
14	122	38	137	75	152	87	166
20	123	51	138	79	153	88	167
26	124	58	139				

APPROVED 29 JAN 1964 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust AS 82	TITLE SETSCREW-HEXAGON SOCKET, FLAT POINT, ALLOY STEEL, CADMIUM PLATED, UNF-3A, PLAIN AND SELF-LOCKING Ⓐ	MILITARY STANDARD MS51966
PROCUREMENT SPECIFICATION FF-S-200	SUPERSEDES: MS51026, MS51028 and in part MS18065, BCTX2.6, BCTX2.7, BCTX2.13, BCTX2.15, BCUX1, BCUX3 and BCUX4.5	SHEET 2 OF 2

DD FORM 672-1 (Cancelled)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
HEADQUARTERS
U. S. ARMY WEAPONS COMMAND
ROCK ISLAND, ILLINOIS 61201

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE, $300



POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Weapons Command
ATTN: [redacted] SWERR-E-SC
Rock Island, Illinois 61201

FOLD

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.